SET: B

AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH

Faculty of Science and Information Technology

Department of Mathematics

MAT 3103 : Statistics and Probability (All Sections)

Mid-term Examination Summer : 2018-19

Total Marks : 40 Time : 2 hours

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Instructions : 1. Answer any 4 (Four) questions .
2. The figures in the margin indicate full marks.

1. **a)** Calculate coefficient of variation for the number of defective devices produced by a firm in different days (x): 12, 16, 11, 11, 10. Comment on the skewness of the data. **5**

b) The following is the distribution of emails received by different organizations:

Class interval of emails	15-20	20-25	25-30	30-35	35-40	Total
No. of organizations, f	20	15	10	20	5	70

Calculate geometric mean and standard deviation of the distribution. **5**

2. **a)** Three fair coins are tossed once. Find the probability that (i) at best two coins will show head, (ii) at least two coins will show head. **3**

b) In a box there are 30 balls; 16 blue and 14 red. 6 blue and 4 red balls are good. One ball is selected randomly. Find the probability that the selected ball is (i) defective, (ii) red or good. **3**

c) In a library, there are 70% science books, 30% arts books. Among science books, 60% are foreign. Among arts books, 40% are foreign. A foreign book is selected randomly. Find the probability that the selected book belongs to science category. **4**

3. **a)** The joint probability density function of two continuous random variables X and Y is given by: $f(xy) = 2x,\ 0 < x < 1,\ 0 < y < 1$. Calculate (i) P ($X > 0.5$) and (ii) V ($3X$). **5**

b) Calculate V (2X - 4) for the given discrete random variable X: **3**

X	0	1	2	3	4	Total
P(X)	0.3	0.2	0.4	0	0.1	1

c) Define random variable with an example. **2**

4. **a)** Given, $M_x(t) = (q + pe^t)^n$. Calculate mean and variance. **4**

b) In an industry, 70% workers get injury during work. 4 workers are selected randomly. Find the probability that (i) 2 workers get injury and (ii) at least 2 worker get injury. **3**

c) If 2% fuses produced by a company are usually found defective, what is the probability that in a box of 200 fuses, there will be (i) no defective fuses, (ii) at most 1 defective fuse. **3**

5. **a)** The joint probability density function of discrete random variables X and Y is given by

X \ Y	0	1	2	3	g(x)
1	0.2	-	0.1	0.1	0.4
2	-	0.3	0.2	0.1	0.6
h(y)	0.2	0.3	0.3	0.2	1.0

Calculate (i) $P(Y \geq 1)$, (i) $E(2X-Y)$ and (iii) $V(2X - 4)$. **6**

b) Given, $M_x(t) = (1-\lambda t)^{-1}$. Calculate mean and variance. **4**